

Rithvik Mysore Suresh

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PROFESSIONAL SUMMARY

Software Engineer with **2+ years at Mercedes-Benz Research & Development India**, delivering **high-scale React system** supporting **5,000+ users** with **99.9% uptime**. Proven experience in **system design, performance optimization**, and **CI/CD automation**, driving measurable improvements in **enterprise-grade production environments**. Strong background in **AI-driven systems**, including **LLM-powered applications, RAG pipelines, vector embeddings**, and **semantic search**, enabling intelligent, data-driven solutions.

EDUCATION

Indiana University Bloomington <i>Master of Science in Computer Science</i>	Aug 2024 – May 2026 <i>Bloomington, IN</i>
Siddaganga Institute of Technology <i>Bachelor of Engineering in Computer Science</i>	Aug 2018 – Jul 2022 <i>Tumakuru, India</i>

WORK EXPERIENCE

Graduate Project Intern <i>IU Capital Planning and Facilities</i>	Oct 2024 – Present <i>Bloomington, IN</i>
Engineered and executed a Python-based analytical and reporting framework to investigate complex data inconsistencies, perform multi-dimensional data validation, and generate high-fidelity capital planning reports.	
Software Specialist <i>Mercedes-Benz Research & Development India</i>	Aug 2022 – Jul 2024 <i>Bengaluru, India</i>
- Contributed significantly to OneMB React-based internal application serving 5,000+ employees; owned performance optimization and system design achieving 99.9% uptime. - Architected and owned end-to-end React.js optimization initiative improving website rendering performance by 30% through lazy loading, code-splitting, and image delivery enhancements; directly impacting 5K+ monthly users - Spearheaded production deployment strategy establishing automated CI/CD pipeline reducing manual effort by 75%; mentored a junior engineer on React best practices, system design patterns, and code quality standards.	
Graduate Engineering Trainee <i>Mercedes-Benz Research & Development India</i>	Apr 2022 – Jul 2022 <i>Bengaluru, India</i>
- Executed production-grade WordPress deployments with <1 minute downtime, adhering to standardized release and rollback procedures to ensure platform stability. - Developed 8+ reusable UI components and micro-frontends using modular design principles, reducing code duplication by 25% and improving long-term maintainability through cross-regional collaboration.	

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C, SQL, HTML, CSS
Frameworks & Libraries: React.js, Node.js, React Native, Flask, Apache Spark, Django
AI & Machine Learning: Claude LLM, Generative AI, RAG (Retrieval Augmented Generation), Vector Embeddings, Semantic Search, NLP, Prompt Engineering, AI System Design, MongoDB Atlas, Vector Search, Elasticsearch
DevOps & Deployment: Git, Docker, Kubernetes, Terraform, CI/CD (GitHub Actions, Vercel, Render), Production Monitoring
Methodologies & Concepts: Object-Oriented Design, Data Structures & Algorithms, Secure Coding, Agile/Scrum, SDLC, AI Integration Patterns, Microservices Architecture, System Design

ACADEMIC PROJECTS

JobMate - Career Guidance App <i>React, Claude LLM, RAG, Machine Learning, Open Source</i>	Jan 2026 – Present
- Built full-stack AI platform with React frontend, Claude LLM engine, vector embedding RAG system, MongoDB backend, and Adzuna API achieving 98% semantic accuracy on 500K+ career-job matching queries. - Developed ML-powered resume analyzer with ATS optimization scoring algorithm, NLP-based skill extraction, real-time job matching, sub-100ms latency processing 500+ daily requests with 99.9% SLA.	
3D World Explorer <i>Swift, SwiftUI, SceneKit, Claude API</i>	Nov 2025 – Dec 2025
- Built iOS app integrating AI to generate custom world descriptions and render real-time 3D visualizations with 60 FPS performance using SceneKit - Designed responsive SwiftUI interface with interactive 3D scenes, smart prompt detection (castle, city, garden, forest, steampunk), and user-friendly text manipulation features	
CuraSure <i>React, JavaScript, MongoDB, Docker, Vercel</i>	Jan 2025 – May 2025
- Engineered a full-stack MERN platform with role-based access control supporting patients, doctors, and providers, enabling secure, scalable, and workflow-driven healthcare operations. - Implemented authentication and 2FA, designed and deployed 25+ RESTful APIs (appointments, insurance, real-time chat), and shipped a production-ready system using Vercel and MongoDB Atlas with CORS-secured endpoints.	